Flight Dynamics of the Schleicher ASK 21 in JSBSim

A physics-based component model checked against the AFFTC spin evaluation and the flight manual, with a comparison to the FlightGear reference model

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Abstract

Two JSBSim flight-dynamics models of the Schleicher ASK 21 two-seat training sailplane are flown through an identical, fully scripted flight-test programme and compared with the behaviour documented in the US Air Force Flight Test Center stall and spin evaluation (AFFTC-TR-89-27), the manufacturer's flight manuals and the published rigging data. The first is the reference model distributed with FlightGear (FGAddon revision 8166), a lumped-coefficient description of Aeromatic origin. The second, called v9 here, is a new component build-up: 22 wing, 6 tailplane and 3 fin strips with NeuralFoil section data over ±180°; a pre-solved lifting-line operator that keeps the JSBSim function graph acyclic; a post-stall model with explicit loss of leading-edge suction; slender-body fuselage loads computed from the three-view; each wing panel's first bending mode, its stiffness set by a ground vibration test of the type; and the AFFTC mass properties. Eighteen post-stall and interference factors that no source provides were identified by CMA-ES against the AFFTC spin-susceptibility map, spin characteristics, recoveries and stall data, within physically motivated bounds, at the 240 Hz rate at which the spins converge. The programme comprises 186 test points per model: glide polar, stall speeds and stall behaviour, all 24 AFFTC loadings with three entry techniques, six recovery inputs at four loadings, roll, sideslip, airbrakes, inverted stall, ground attitude, a hands-free aerotow, small-perturbation modes, a phugoid speed sweep and robustness. The reference model reproduces the polar to 0.09–0.17 m/s rms with a maximum glide ratio of 42 (manual: 33.5), has its neutral point at 16.5 in, forward of the aft CG limit, and does not spin at any loading: with pro-spin controls it enters a spiral dive. v9 reproduces the polar to 0.025–0.030 m/s rms and the manual's stall speeds within a few km/h, and matches the AFFTC classification at 20 of 24 loadings. Its developed spin is oscillatory at 79–128 deg/s, and it recovers by the manual procedure in less than one turn. Its deficiencies are quantified: a phugoid that diverges below about 115 km/h, height loss per spin turn (291 vs. 200 ft), spin pitch-oscillation amplitude, the aft-CG g-break and the nose drop in dynamic stalls. A force-balance argument shows that the AFFTC height-loss figure implies a vertical force coefficient at or above the two-dimensional flat-plate limit. This third issue adds the wing's bending, moves the model to 240 Hz and re-identifies the post-stall factors; Section 9 lists the changes since the first issue.

1 Introduction

The ASK 21 is the standard two-seat trainer of many gliding clubs and, with its tail-ballast kit, widely used for spin training. A simulator model of it is therefore expected to do two things well that a lumped-coefficient model of a cruising aircraft rarely does: reproduce the manufacturer's performance and low-speed handling, and depart, spin and recover in the way the real glider does, as a function of centre of gravity (CG) *and* of the inertia added by tail ballast. The most complete public description of the latter is the US Air Force Flight Test Center (AFFTC) stall and spin evaluation of 1989 [1], flown after a fatal USAF Academy TG-9 (ASK 21) accident in November 1988, with 24 documented loadings, mass properties, and spin-susceptibility boundaries in the CG–pitch-inertia plane.

This report compares two JSBSim [9] models against that evidence and against the flight manuals [2, 3]:

- **FlightGear reference model** (FG): `ask21-jsb.xml` and `Systems/ask21-aero.xml` of FGAddon revision 8166, an Aeromatic-generated [13] lumped model with a tabulated $C_L(\alpha)$ to ±25° and constant stability derivatives;
- **v9**: a component build-up written for this study and generated by a script from geometry, airfoil and mass data, with flexible wings. It keeps the FlightGear interfaces, so it is a drop-in replacement.

Both models are flown by the same scripted test pilot through 186 test points each (Section 6). All result tables except the summary scorecard are written directly by the evaluation scripts; the scorecard quotes their values.

2 Reference data

Table 1 lists the sources. Three points of data reduction deserve mention. *(i)* The OCR text of AFFTC Table C1 lost the sign of the rear-seat arm. With −3.15 in instead of +3.15 in the report's CG column is reproduced to within 0.11 in for 23 of the 24 loadings (the remaining one, 15.00 in, comes out 0.33 in aft; its listed gross weight is also 4 lb heavier than its listed ballast implies). The systematic +0.1 in with tail ballast stems from a 1 lb mounting bracket assumed for the tail weights. The report datum, the wing-root leading edge, lies at $x = -1.637$ m in the JSBSim structural frame shared by both models, which reproduces the reported empty CG of 29.02 in. *(ii)* The AFFTC spin boundaries (Fig. 5 of [1]) are linear in the CG–I_{yy} plane. Through the report's quoted values (incipient spins aft of 13.0 in at low and 12.5 in at high inertia, sustained aft of 15.0 and 13.5 in), taking low and high inertia as 7.8 and 9.1×10^6 lb in^2, they read

$$x_{\mathrm{inc}}(I) = 13.0 - \tfrac{0.5}{1.3}\,(I - 7.8)\ \text{in}, \tag{1}$$

$$x_{\mathrm{sus}}(I) = 15.0 - \tfrac{1.5}{1.3}\,(I - 7.8)\ \text{in}. \tag{2}$$

A loading forward of x_{inc} does not spin; one aft of x_{sus} sustains a spin indefinitely with pro-spin controls, and one in between produces only incipient spins (at least one turn, then self-recovery). *(iii)* All AFFTC airspeeds are *indicated*. The report stresses that the pitot–static system read zero or less in large sideslip and was unreliable in spins. The models report calibrated airspeed (CAS); in spins and large slips the AFFTC indications read low or zero.

Table 1: Sources and what they contribute.

Source	Content used
AFFTC-TR-89-27 [1]	24 loadings (Table A1: CG, pilot and ballast masses, I_{yy}); empty inertias (App. C); moment arms; spin boundaries (Fig. 5); stall, spin, recovery, inverted and sideslip observations
Flight manual TM 04b [2]	stall speeds with and without airbrakes; airbrake dive speed; spin recovery; sideslip remarks
Flight manual, English edition [5]	rigging data (p. 54); flight polar (p. 55); aerobatic entry speeds
Flight manual TM 23 [3]	aerotow and winch procedures ("trim neutral", "take-off at about 75 km/h")
ASK 21 Mi flight manual [4]	tailplane, fin, rudder, elevator and airbrake geometry; fuselage surface area
Three-view [6]	planform, dihedral, fuselage contours, wheels, root-chord angle
UIUC data base [7]	FX S 02-196, FX 60-126, FX 71-L-150/30 coordinates
NeuralFoil [8]	2-D section c_l, c_d, c_m incl. flap deflections, $-25^\circ \ldots 30^\circ$
OSTIV 2010 study [14]	winch-launch force schedule
KTH ground vibration test [10, 11]	first symmetric and antisymmetric wing bending of the glider on soft springs (2.95 and 6.50 Hz); the wing shell's lay-up
EASA TCDS A.221 [12]	maximum mass and maximum mass of the non-lifting parts (the wing panels' mass)
Informal	a flight video (first 180° of a full-aileron roll in $\approx$ 5 s); the commissioning glider pilot's experience: hands-free lift-off before the tug, a full slip at least as effective as the airbrakes

3 Airframe, geometry and mass properties

The planform was digitised from the three-view at 1500 units/m: 17.0 m span, straight leading edge, double-tapered trailing edge, 4° dihedral, aileron from 5.20 to 7.95 m semispan, and an airfoil change from FX S 02-196 at the kink to FX 60-126 at the tip (Fig. 1). The rigging checks of the flight manual [5] (45, 36 and 13 mm over 1470, 1000 and 590 mm gauges) set the root chord 1.75° above the tailplane, with +0.31° wash-in at the kink, −0.49° washout at the aileron's outer end and −0.6° extrapolated to the tip. The root chord lies at 2.48° to the body x-axis of the JSBSim/FlightGear frame (2.6° to the horizontal of the three-view), so the tailplane sits at +0.73°. *Ground attitude.* The manufacturer's rigging drawing [5], drawn in the levelled attitude, shows the nose- and main-wheel bottoms level to within 1 cm over their 2.25 m spacing. In the body frame this puts the nose-wheel contact 1.9 cm above the main-wheel contact; the FlightGear gear and 3-D model have it 7.3 cm above, which tilts the glider about 1.4° further nose-down on its wheels. v9 uses the corrected contact (the 3-D model's nose wheel is lowered by the same 5.4 cm), so it rests at 0.5–0.6° pitch instead of −0.9 to −0.8°, with the wing root about 3° instead of 1.7° to the runway. The independent three-view [6] implies 4.3° and is the outlier. Integrating the fuselage contours of the side, top and front views with a super-elliptic section factor of 0.82 gives length 8.38 m, volume 2.18 m^3, and wetted area 13.5 m^2. The skin-friction build-up uses the manufacturer's 12.33 m^2. Control travels follow from the rigging data: aileron +10.8/−26.5°, elevator +16.2/−22.6°, rudder ±27.8°.

Table 2: Mass properties: empty-aircraft inertias and agreement with the 24 AFFTC loadings (each model's own moment arms).

Quantity	AFFTC App. C	FlightGear	v9
I_{xx}, 10^6 lb in^2 (empty)	30.2	4.63	30.20
I_{yy} (empty)	6.7	1.39	6.70
I_{zz} (empty)	34.3	10.91	34.30
I_{yy} over the 24 Table A1 loadings, rms error	–	5.68 (−67 %)	0.05 (−0 %)
CG over the 24 loadings, max. deviation (in)	–	0.80	0.33

v9 places the occupants and ballast at the AFFTC arms (front seat −47.91 in, rear seat −3.15 in, tail ballast 209.8 in, cockpit ballast −63.39 in) and uses the measured empty inertias. The test-programme variant carries the test glider's 851 lb empty weight. FG keeps Aeromatic estimates that are smaller by factors of 3.1 to 6.5 (Table 2, Fig. 2). The AFFTC found spin susceptibility to depend on I_{yy} as well as on CG (1–2), so FG starts with a handicap for the spin map. Together with its weaker roll damping, its small I_{xx} also shortens the roll time constant fourfold (Table 12).

4 The v9 aerodynamic model

All forces are computed per element in body axes from the local air velocity

$$\mathbf{v}_i = \mathbf{v}_{\text{air}} + \boldsymbol{\omega} \times (\mathbf{r}_i - \mathbf{r}_{\text{CG}}), \tag{3}$$

summed, and referred to the aerodynamic reference point. JSBSim then transfers them to the current CG. Rotation about the CG therefore enters every strip exactly, which is the essential ingredient for spins and rolls at large rates.

4.1 Wing: strip theory with an explicit lifting line

Each of the 22 strips k has a geometric section angle $\alpha_{g,k}$ from the flow at its 3/4-chord point, rotated into the dihedral plane, plus the local incidence. Induced angles follow from a Weissinger-type influence matrix F (trailing vortex sheets of all strips at the collocation points), with $\Gamma_j = \tfrac{1}{2} c_j V_j c_{l,j}$ and $\alpha_{i,k} = \sum_j F_{kj}\Gamma_j / V_k$. JSBSim's function graph must be acyclic, so the implicit system cannot be iterated. It is linearised about the section slope a and solved once, offline:

$$\begin{aligned} \boldsymbol{\alpha}_i &= (\mathsf{I} + a\,\mathsf{F}_c)^{-1} \mathsf{F}_c \left[c_l(\alpha_g)\, V \right] / V, \\ \mathsf{F}_c &= \tfrac{1}{2}\, k_{\text{ind}}\, \mathsf{F}\, \mathrm{diag}(c). \end{aligned} \tag{4}$$

The nonlinear $c_l(\alpha_g)$ inside the bracket keeps the operator consistent at moderate stall, and a fade to zero between $|\alpha_g| = 25^\circ$ and 70° removes it in fully separated flow. With $a = 6.7\,\text{rad}^{-1}$ the 22-strip operator gives $C_{L_\alpha} = 5.95\,\text{rad}^{-1}$ and a Trefftz-plane efficiency $e = 1.06$. The value above one is a discretisation artefact, removed by $k_{\text{ind}} = 1.07$ (converged strip count). Ground effect multiplies $\boldsymbol{\alpha}_i$ by $\phi = (16h/b)^2/[1 + (16h/b)^2]$ [15]. The resulting section force is normal and tangential to the local velocity *at the bound vortex* (1/4 chord, Kutta–Joukowski), whereas the 3/4-chord flow only sets its magnitude. The two differ by $qc/2V$ under a pitch rate (Section 9).

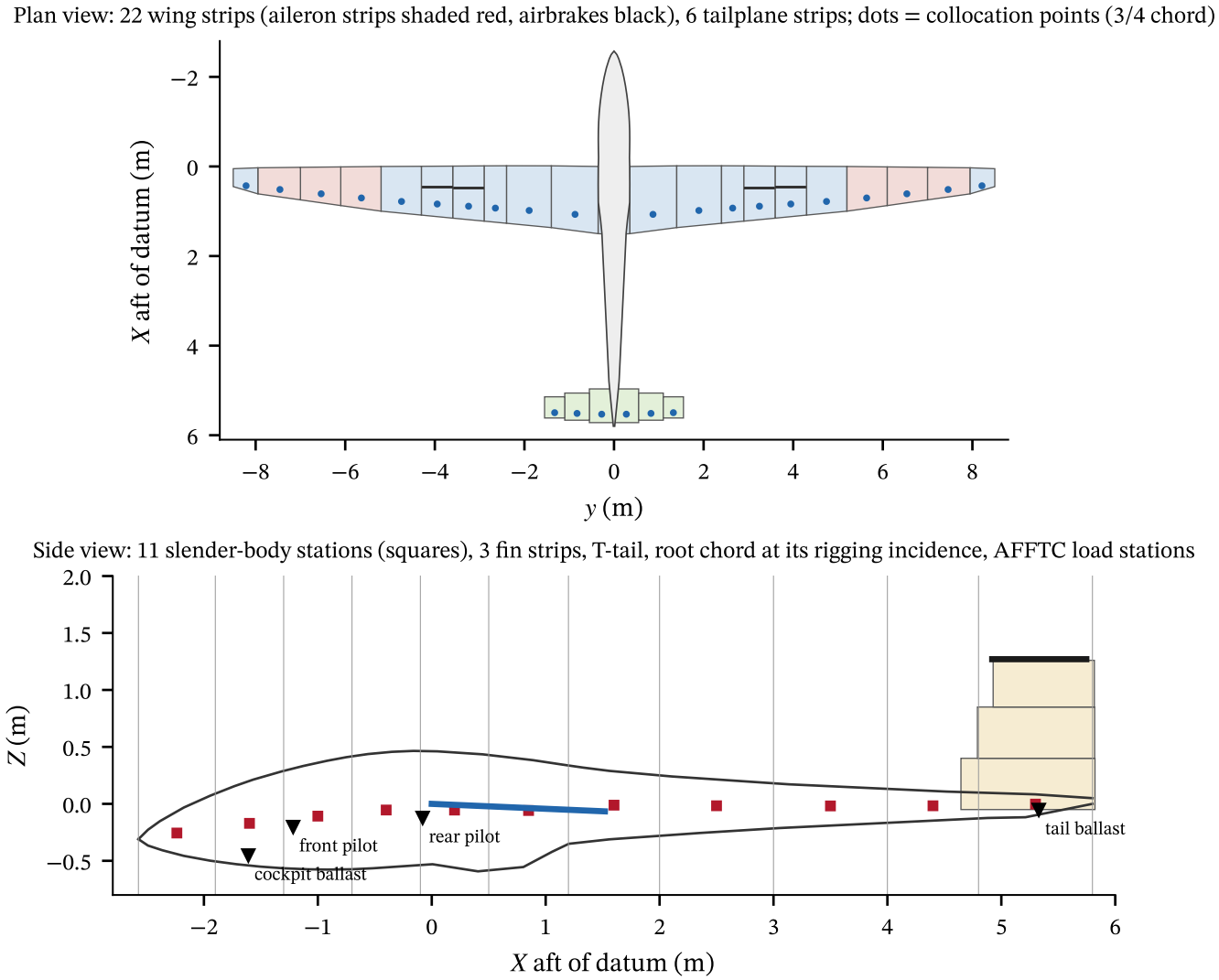


Figure 1: v9 discretisation drawn from the generator's data. Wing chords, leading-edge line and dihedral come from the three-view, and are scaled to the type-certificate area of 17.95 m^2 (the drawing integrates to 18.06 m^2). Collocation (3/4-chord) points carry the lift magnitude; forces act at the 1/4-chord points. Fuselage stations carry slender-body loads. Triangles mark the AFFTC load stations.

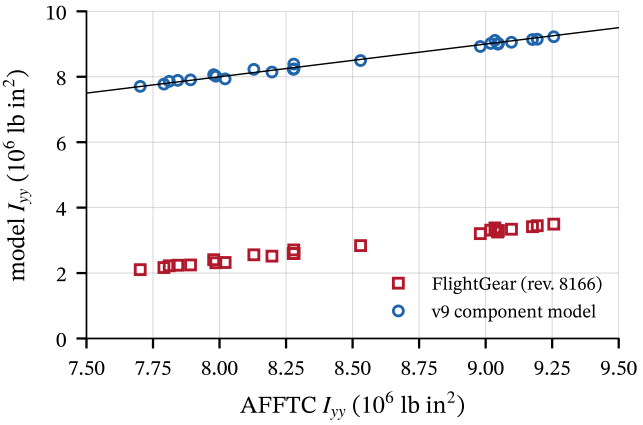


Figure 2: Pitch inertia of the 24 AFFTC loadings: model vs. measured.

4.2 Wing bending

Each wing panel bends in its first mode, one degree of freedom per panel: the tip deflection q. The panel is a beam clamped in the fuselage, 95 kg (the type certificate's 600 kg less its 410 kg of non-lifting parts [12]) distributed with the square of the chord. Its stiffness is the spar's, sized to one strain allowable at the aerobatic limit load (+6.5 g, 600 kg), $EI = M_{\lim} h / 2\varepsilon$ with the spar depth $h = 0.9\,t$, plus the shell's glass faces (0.78 mm of ±45° fabric per surface over 80 % of the chord [11]), which carry the outer wing where the spar runs out. The strain level is not taken from any source but from the measured first symmetric wing bending of the whole glider hanging on soft springs, 2.95 Hz [10]: with the rest of the 380 kg test glider as a rigid mass at the roots it requires $\varepsilon = 0.92\,\%$. The same beam then gives 6.0 Hz for the first antisymmetric bending against the test's 6.50 Hz (which also contains the fuselage's bending), and 2.6 Hz with the fuselage held; without the shell it would be 5.2 Hz. Structural damping is 2 %. In level flight at 500 kg the tips rise 0.17 m; on the ground the wing sags 8 cm.

The mode is driven by the generalized force of the strips' normal forces and by the panel's inertia in the accelerating airframe (load factor, roll and pitch acceleration, the centrifugal terms of the body rates), and integrated semi-implicitly, with the static deflection at an initial condition. Each strip moves with it: it rises by $\phi(y)q$, its dihedral grows by the mode's slope, and its section sees the plunge velocity $\phi(y)\dot{q}$. The airframe feels the panels' acceleration at the mode's centroid. The tip skids (and the ground-handling wing supports) follow the tip, and the contact with the ground acts on the mode im-

plicitly, so that the stiff skid does not limit the time step. The bending raises the effective dihedral: C_{l_β} is -0.115 instead of -0.085 for the rigid wing, and the 45–45° roll reversal is about 0.3 s slower.

4.3 Sections

Pre-stall data come from NeuralFoil ("xxlarge") at the strip Reynolds number, with the profile drag scaled as $c_d \propto Re^{-0.22}$ for off-design speed. Above $0.8\,c_{l,\max}$ the lift follows a monotone trailing-edge-stall shape to a peak $k\,c_{l,\max}^{\mathrm{NF}}$ at α_{peak}, then drops by Δc_l over 5°. The lift removed by early separation reappears as pressure drag. Beyond $\alpha_{\mathrm{peak}} + b_{\mathrm{off}}$ the data blend over d_{stall} degrees into a flat plate,

$$\begin{aligned} c_N &= C_{N,90}\sin(\alpha + k_\delta\delta\cos^2\alpha),\\ x_{cp} &= 0.25 + (x_{cp,90} - 0.25)\,|\alpha|/90^\circ, \end{aligned} \tag{5}$$

with reverse-flow handling up to ±180° (Fig. 3). The decisive addition for departure behaviour is the *chordwise* force. Attached flow carries leading-edge suction, $c_A < 0$. Once the section has stalled, c_A is blended within w_{ca} degrees from its value at the lift peak to a small positive value, while c_N is kept:

$$\begin{aligned} c_A(\alpha) &= (1-s)\,c_A(\alpha_{\mathrm{pk}}) + s\,\max\big(c_A,\ c_{A,\mathrm{sep}}\cos\alpha\big),\\ s &= S\big((\alpha-\alpha_{\mathrm{pk}})/w_{ca}\big), \end{aligned} \tag{6}$$

where S is a smoothstep. Without this term a stalling wing that drops keeps pulling itself forward and yaws *away* from the drop; with it, the dropping wing's force tilts aft and yaws it into the spin, as observed.

4.4 Controls and airbrakes

Ailerons, elevator and rudder are plain flaps evaluated by NeuralFoil on the deflected geometry. Beyond ±12° NeuralFoil saturates, so the increment curve of the reference deflection is scaled by the increment at $\alpha = 0$. A Datcom-type effectiveness factor (fitted: 1.01) accounts for gaps and seals. The stick-centred elevator angle is not documented. A linear linkage would give the mid-travel value of −3.2°. The adopted −6° makes the stick-free, trim-neutral aerotow take-off (manual: "trim neutral", "take off at about 75 km/h") lift off at 80 km/h with two 80 kg pilots (Table 11). The airbrakes act as section increments $\Delta c_l, \Delta c_d, \Delta c_m$ over 2.9–4.3 m semispan, with Δc_d calibrated to the manual's 232 km/h in a 45° dive (flight path 45° below the horizon) at 600 kg.

4.5 Tailplane and fin

The T-tail sees the wing's downwash from the same trailing-vortex model, tabulated against the tail's height above the convected wake sheet (inclined at $\alpha - \varepsilon$, $\varepsilon \approx 0.28(\alpha + 7^\circ)$, $\mathrm{d}\varepsilon/\mathrm{d}\alpha = 0.28$ at the tail). It includes a lag l_t/V through $\dot\alpha$, an image-vortex ground-effect factor, and a Silverstein-type Gaussian wake deficit

$$\begin{aligned} \frac{\Delta q}{q} &= k_w\,\frac{2.42\sqrt{c_d}}{x/c + 0.3}\,\exp\Big[-0.69\,\frac{h^2}{c_d\,b_w^2}\Big],\\ b_w &= 0.68\,c\sqrt{x/c + 0.15}. \end{aligned} \tag{7}$$

Tailplane and fin induced angles come from one-mode lifting lines (fin effective aspect ratio 2.35 with the T-tail end plate). The fin additionally loses dynamic pressure in the separated cross-flow wake of the fuselage at large local sideslip.

4.6 Fuselage and drag build-up

Eleven stations carry slender-body loads, Munk's potential term plus Allen–Perkins viscous cross-flow [16]. In the vertical plane a station carries

$$F_z = -k\,\rho\,u\,w\,\Delta S - \tfrac{1}{2}\rho\,c_{dc}\,A_p\,w\,V_c, \qquad V_c = \sqrt{v^2 + w^2}, \tag{8}$$

and correspondingly F_y with v in the lateral plane; u, v, w are the local velocity components, ΔS the change of the equivalent section area across the station and A_p its projected area. The potential term equals $q\,\Delta S\sin 2\alpha_t$ resolved into the cross-flow plane. The wing's up- and downwash is added to w ($k = k_2 - k_1 = 0.94$ ahead of the maximum section, $k\,k_{\mathrm{aft}}$ behind it). Zero-lift drag is $C_f\,FF\,S_{\mathrm{wet}}$ for the fuselage ($C_f = 0.0026$, $FF = 1.09$), a gear drag area of $0.010\,\mathrm{m}^2$, and 0.0017 miscellaneous C_D.

4.7 Implementation and parameter identification

The generator writes about 700 JSBSim functions (713 kB) and two systems, the wing bending and the wake of a tug ahead. In FlightGear 2024 every property read takes a lock, so the graph is laid out to read few: each station's velocity is the velocity at the structural origin (worked out once per step) plus the rotation times its fixed position, the moments are taken about that origin and moved to the reference point once, on the force sums, the lifting line works on the sums and differences of the mirror-image strips, functions read in only one place are written into their reader, and the tug's wake runs in a channel that is switched off when there is none. That halves the reads per step to about 3500. The model runs at 50 μs per step offline, 83 times real time at 240 Hz, against 4 μs for FG.

Rate. At 120 Hz, FlightGear's default, 8 of the 72 spin entries of the test programme (Section 6) end in a different class than at 240 Hz; at 480 Hz one does. The borderline spins are sensitive to the integration error, everything else changes by less than 1 %. v9 therefore runs at 240 Hz, set in its FlightGear aircraft file, and was identified there.

Eighteen factors without a source (Table 3) were identified with CMA-ES [17], population 12. The objective sums (a) spin-map misclassifications, softened by the distance to the AFFTC boundary; (b) deviations of the developed-spin rate, height loss, pitch attitude and its oscillation, airspeed, bank and load factor from the AFFTC ranges; (c) recovery turns for six inputs at four loadings; (d) 1-g stall character at four CGs (elevator limit, g-break, bucking, wing drop); and (e) the manual's stall speeds. Stage 1 (30 generations, stick-centred elevator at mid travel) was followed by a local stage (12 generations, elevator −6°, spin-rate and height-loss weights tripled). After the corrections of the second issue the parameters were re-identified in a third local stage of 12 generations from the stage-2 result (rigid wing, 120 Hz; objective 12.59). The bending changed the departures enough to drop the spin map from 22 to 17 matching loadings, so a fourth local stage of 20 generations, with the bending and at 240 Hz,

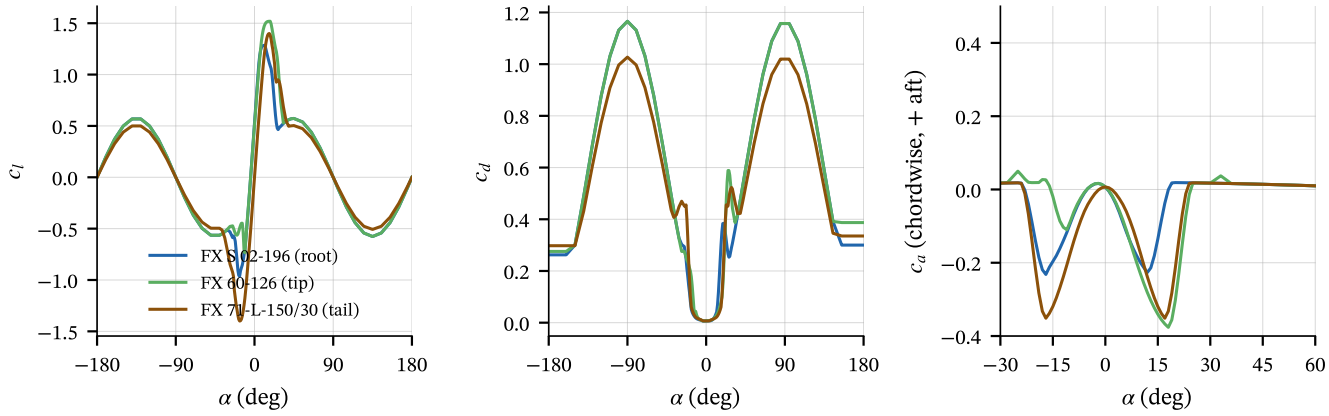


Figure 3: v9 section data (fitted parameters) over ±180°, and the chordwise force c_A with the collapse of leading-edge suction after the stall (right).

started from that result (objective 19.12 there). Its best candidate (objective 12.14, 4 misclassifications) is the present set; two factors ended at their bounds, $x_{cp,90}$ and the root section's α_{peak}. Each objective evaluation flies 100 dynamic test points (about 30 s at 240 Hz on 8 cores).

5 The FlightGear reference model

FG is a classical build-up about a fixed aerodynamic reference point. $C_L(\alpha)$ is tabulated from −25° to +25° (peak 1.51 at 12°) and held constant outside, with a C 172-derived ground-effect table. $C_D = C_{D0}(\alpha) + 0.018\,C_L^2$ plus sideslip, elevator and airbrake terms; damping and cross derivatives are constants ($C_{m_q} = -13.5$, $C_{m_{\dot\alpha}} = -12$, $C_{l_p} = -0.4$, $C_{n_r} = -0.15$, $C_{l_r} = 0.15$, $C_{n_\beta} = 0.15$, $C_{l_\beta} = -0.1$, all per rad on its own reference lengths). Effectiveness tables reduce aileron authority beyond $\alpha = 15°$ and elevator authority beyond 20°. There is no stall asymmetry, no rate dependence of the stall, and no α dependence of the rotary derivatives, and the inertias are those of Aeromatic (Table 2).

6 Test method

Both models are run in JSBSim 1.3 (Python bindings) on the same aircraft root, each at its rate in FlightGear: v9 at 240 Hz, FG at FlightGear's default 120 Hz. Only the aerodynamics, the wing-bending system and the aircraft file differ; the FCS system file is identical. Loadings are defined by the AFFTC pilot, tail-ballast and cockpit-ballast masses. FG has no cockpit-ballast station, so its cockpit ballast is added to the front pilot.

Static evaluation. Static coefficients and trims use the full JSBSim model with frozen integrators. JSBSim derives $\dot\alpha$ from the instantaneous accelerations, which are non-zero at an unbalanced static point. For coefficient curves and derivatives, each model's $\dot\alpha$ input is therefore replaced by a free property that is set explicitly.

Pilot models. Proportional–integral attitude and airspeed loops with rate feedback. Wings are held level with aileron and the ball centred with rudder unless the test prescribes otherwise.

Stall and spin entries as flown by the AFFTC: *WL*, the nose is held 10° high until the stall, then full aft stick and full rudder, aileron neutral; *WL+A*, the same with full aileron into the spin; *TH*, the "thermal entry": full aileron without rudder near the stall, a 1.5 s pause, then full rudder and aft stick, aileron neutral. The AFFTC found the last the most repeatable. Pro-spin controls are held for 40 s.

Classification. Spinning means $\alpha > 25°$ and $|\dot\psi| > 30$ deg/s. A loading is *incipient* if it accumulates at least one spinning turn, and *sustained* if it accumulates at least five and is still spinning in the last 8 s. Its class is the most spin-prone of its three entries.

Recoveries start 14 s after the stall from a WL entry. "Stopped" means a rotation rate below 25 deg/s with $\alpha < 16°$ for 1 s; turns and height loss are counted from the recovery input.

Other tests. The polar is trimmed from 280 km/h down to the stall, each trim starting from the last one, so that none settles on the stalled branch. The 45° airbrake dive is a steady glide along a 45° flight path at 600 kg. The 1-g stall speed is the lowest airspeed before the break. The sideslip is compared with a trimmed full-airbrake glide at the same airspeed. Ground attitude is taken after 10 s at rest; the parked-in-wind test applies a step wind of 15, 25 or 35 kt from five directions for 30 s. The phugoid is excited by a 1.5 m/s speed disturbance with the stick fixed, at six speeds from 80 to 140 km/h.

Small-perturbation modes. Dimensional body-axis derivatives come from central differences of the model's own forces about a trimmed 90 km/h glide, with $\dot\alpha$ as a separate input. The resulting state matrices give phugoid damping ratios of 0.045 (FG) and −0.020 (v9), against +0.076 and −0.019 from the nonlinear simulation at the same speed.

7 Results

7.1 Static aerodynamics

The two models agree on the attached-flow lift slope (6.9 vs. 6.3 rad^{-1} about the respective body axes) but differ in every respect that matters near and beyond the stall (Figs. 4, 5). v9's

Table 3: Identified parameters of v9: bounds and fitted values.

Parameter	Meaning	bounds	fitted
`CN90_wing`	wing section normal force at 90° (flat plate, AR-corrected)	1.1–1.9	1.15
`stall_drop`	lift drop after $c_{l,\max}$	0.03–0.35	0.176
`boff`	onset of the flat-plate blend beyond the peak (deg)	3–12	6.61
`dstall`	width of the flat-plate blend (deg)	8–25	8.55
`w_ca`	width of the leading-edge-suction collapse (deg)	4–18	7.27
`xcp90`	section centre of pressure at 90° (x/c)	0.42–0.56	0.421
`CN90_ht`	tailplane $c_{N,90}$	0.9–1.5	1.02
`CN90_vt`	fin $c_{N,90}$	0.9–1.4	1.12
`wake_k`	wing-wake q-deficit at the tailplane (scale)	0.3–1.8	0.357
`wake_kv`	wing-wake q-deficit at the fin (scale)	0–1	0.33
`fin_blank`	fin q loss in the fuselage cross-flow wake	0.25–0.45	0.287
`cdc`	fuselage cross-flow drag coefficient	0.6–1.3	1.19
`kpot_aft`	aft-body share of the Munk potential load	0.2–1	0.517
`wing_clmax_k`	inner-wing $c_{l,\max}$ / NeuralFoil	0.8–0.9	0.816
`a_peak_root`	inner-wing stall angle (deg)	12–17	12
`tip_clmax_k`	outer-wing $c_{l,\max}$ / NeuralFoil	0.85–1	0.906
`a_peak_tip`	outer-wing stall angle (deg)	15–19	18.8
`flap_eff`	plain-flap effectiveness / NeuralFoil	0.85–1.05	1.01

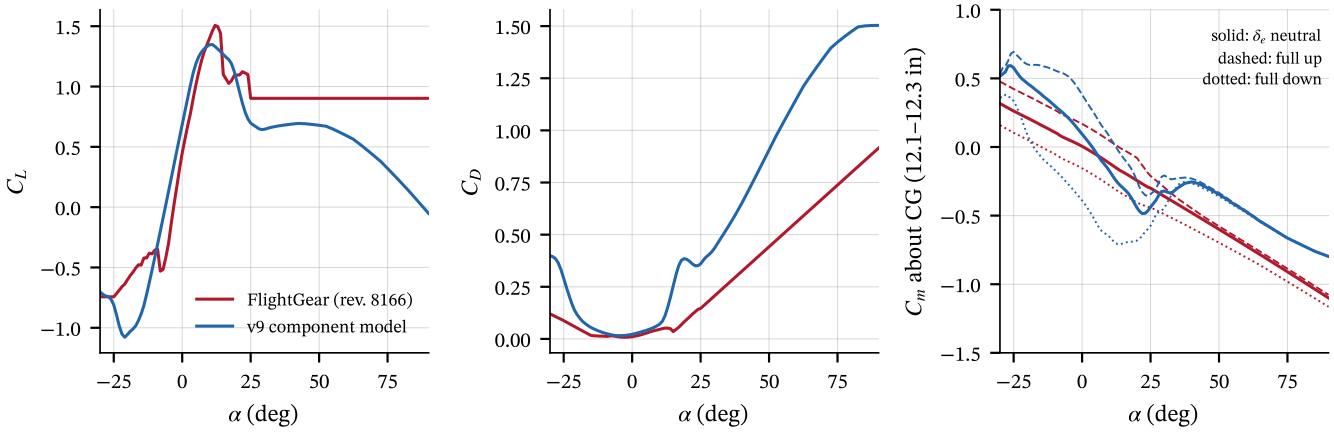


Figure 4: Whole-aircraft static coefficients at 25 m/s, 2 × 80 kg, $\dot{\alpha}$ suppressed. FG holds $C_L = 0.90$ beyond 25° and has a linear $C_m(\alpha)$ without a stall break; v9 shows the stall break, the post-stall normal force, and the loss of elevator power beyond ≈ 35°.

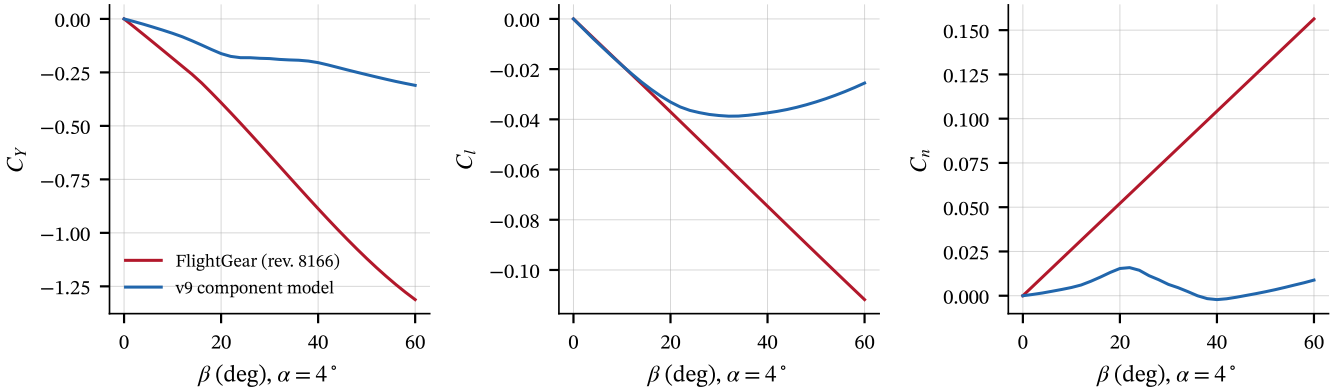


Figure 5: Lateral–directional static coefficients versus sideslip at $\alpha = 4°$. FG's are linear in β to 60°. v9's yawing moment peaks near 22°, where the fin stalls and the fuselage wake starts to blank it, which is what allows the large full-rudder sideslip of Table 10.

stall is gentle ($C_{L,\max}$ 1.35 at 11°, falling to 0.65 at 30° before the normal-force regime) and its pitching moment shows the break that produces the AFFTC's "very mild g-break". FG's pitching moment has no stall break.

7.2 Performance

v9 reproduces the manual polar to 0.025–0.030 m/s rms at the two masses (Fig. 6, Table 4), with a best glide of 32.6–32.7 against 33.5 quoted on the polar and 32.6–33.4 read from its points. FG glides at 42.3–42.5, about 25 % better than the real glider, with its minimum sink at 88–105 km/h instead of 68–72 km/h; its induced-drag factor $K = 0.018$ corresponds

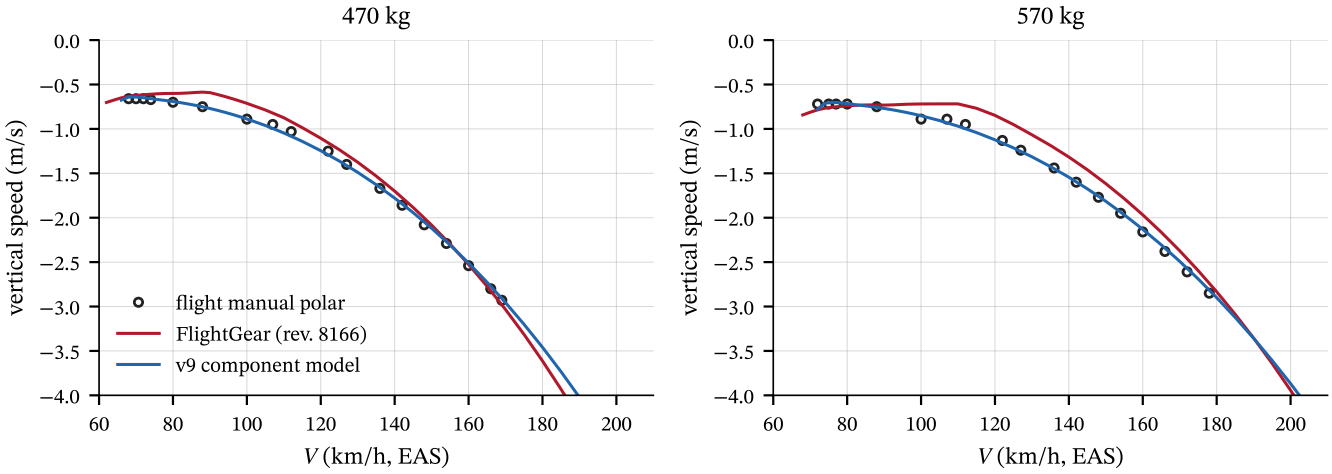


Figure 6: Glide polars against the flight manual (p. 55). Both models are trimmed with the elevator at sea level; the curves are scaled to the exact manual masses.

Table 4: Performance against the flight manual. The 45° dive is flown along a 45° flight path at 600 kg.

	Flight manual	FlightGear	v9
470 kg			
rms sink error (m/s)	–	0.093	0.025
$(L/D)_{max}$ at km/h	32.6 at 88	42.3 at 90	32.6 at 86
min. sink (m/s) at km/h	0.66 at 68	0.59 at 88	0.64 at 70
570 kg			
rms sink error (m/s)	–	0.170	0.030
$(L/D)_{max}$ at km/h	33.4 at 107	42.5 at 110	32.7 at 96
min. sink (m/s) at km/h	0.72 at 72	0.72 at 105	0.70 at 76
V_S single, 470 kg (km/h)	65	60.3	63.9
V_S dual, 600 kg (km/h)	74	69.3	72.8
ΔV_S airbrakes open (km/h)	+3	+1.0 / +1.1	+2.4 / +2.7
45° dive (flight path), airbrakes open, 600 kg (km/h CAS)	232	194	232

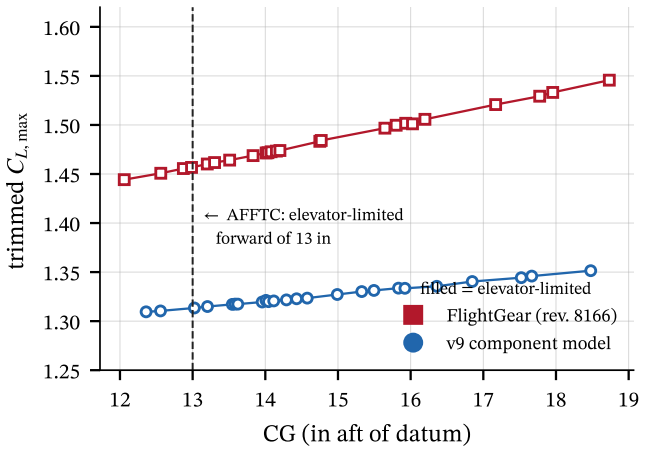


Figure 7: Maximum trimmed lift coefficient over the AFFTC loadings.

to a span efficiency of 1.10. The manual's stall speeds of 65 and 74 km/h come out at 63.9 and 72.8 km/h for v9 and 60.3 and 69.3 km/h for FG. With the airbrakes open the stall speed rises by 2.4–2.7 km/h in v9 and 1.0–1.1 km/h in FG (manual 3 km/h, AFFTC "2 KIAS"). In a 45° dive with the airbrakes open v9 reaches 232 km/h and FG 194 km/h against the manual's 232 km/h; for v9 this is a calibration (the airbrake drag increment was set to it), not an independent check.

Table 5: 1-g stalls at four AFFTC loadings. V_{min} = lowest calibrated airspeed before the break (kt); g-break = pitch-attitude drop within 4 s after the α peak; bucking = peak-to-peak pitch oscillation with the stick held aft.

	FlightGear			v9		
AFFTC loading	V_{min}	g-break	bucking	V_{min}	g-break	bucking
CG 12.36 in (1257 lb)	36.2	0.8°*	2°	38.0	0.8°*	3°
CG 14.01 in (1051 lb)	32.7	4.0°	17°	34.5	1.1°*	7°
CG 15.83 in (1027 lb)	32.1	0.0°	9°	33.0	4.7°	18°
CG 17.67 in (1037 lb)	31.9	0.0°	14°	32.6	8.0°	19°

AFFTC: 33–38 KIAS; forward of 13 in full aft stick before the stall, no g-break (* = elevator-limited); aft of 13 in a very mild g-break of 2–3°, then pitch bucking with the stick held aft.

Table 6: Dynamic (30° nose-high) and inverted stalls; values at the AFFTC loadings 12.36 / 15.83 in. Height loss with the scripted recovery (stick released, pull to level flight above 45 kt).

	AFFTC	FlightGear	v9
30° nose-high entry: min. speed (KIAS)	"as low as 20"	28 / 21	27 / 20
nose drop at the break (deg)	up to 40	0 / 55	65 / 78
height loss (ft)	≈ 100	149 / 167	166 / 242
−1 g inverted stall (KIAS)	38–40	54.4	40.6

7.3 Longitudinal stability and the elevator limit

The stick-fixed neutral point of v9 is at 22.1 in, behind the aft CG limit of 18.46 in. That of FG is at 16.5 in (Table 12): FG is statically unstable over the aft 2 in of the approved CG range, which includes light solo pilots and 4 of the 24 AFFTC loadings. In FG the lift acts at a fixed reference point and only a constant C_{m_α} is added, so its neutral point is fixed by construction rather than by the tail. The AFFTC reports that forward of ≈ 13 in full aft stick is reached before the stall. Neither model reproduces this statically (Fig. 7). In the dynamic 1-g stall at the most forward loading both become elevator-limited (Table 5): FG already needs 89 % of its up-elevator travel for a static trim at $C_{L,max}$, and the dynamic entry uses up the rest; v9 reaches full aft stick and drops its nose by only 0.8°, and does the same at 14.01 in.

7.4 Stalls

The lowest speeds before the break, 32.6–38.0 kt for v9 and 31.9–36.2 kt for FG, lie at or slightly below the AFFTC's 33–38 KIAS (Fig. 8, Table 5). Aft of about 15 in v9 shows the

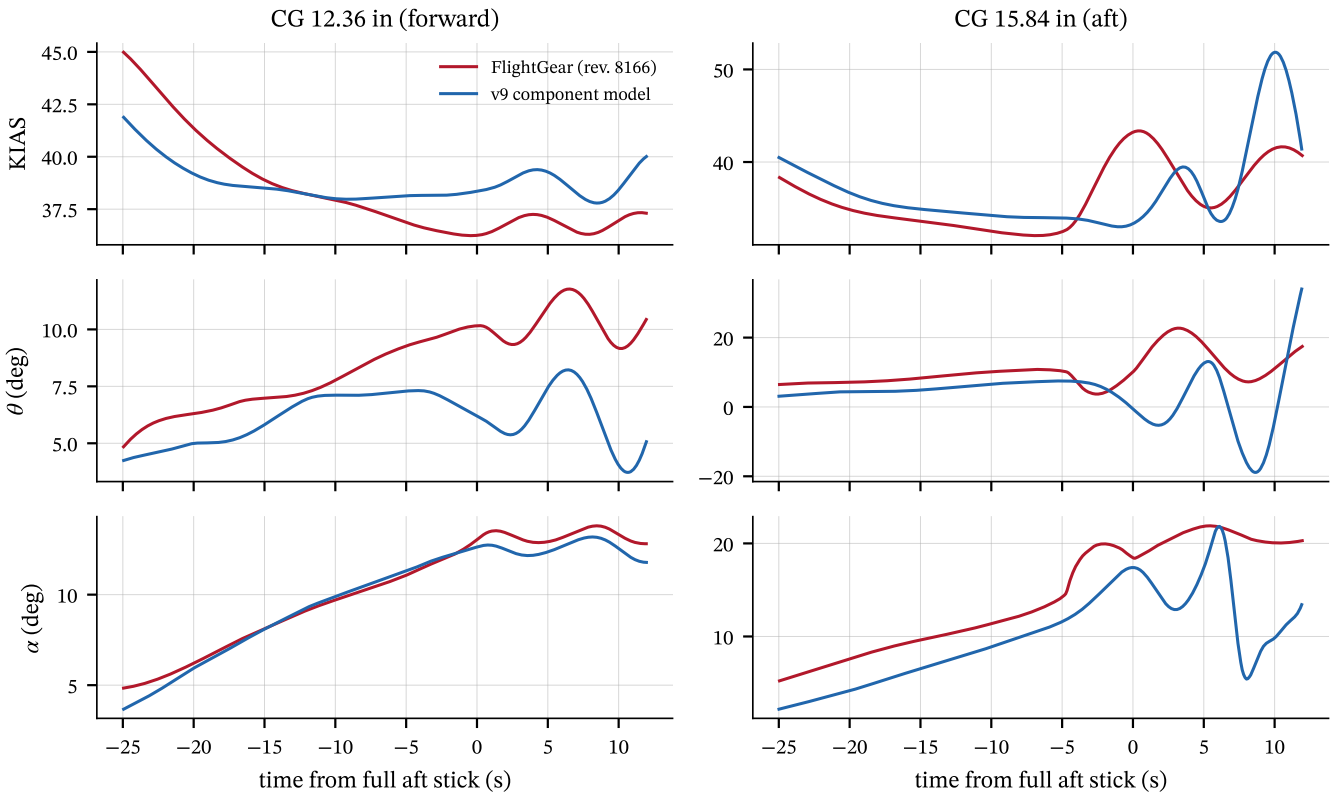


Figure 8: 1-g wings-level stalls (deceleration 1 kt/s, stick then held fully aft for 12 s) at forward and aft CG.

Table 7: v9 loadings classified differently from the AFFTC.

CG (in)	I_{yy}	AFFTC	v9	distance to boundary (in)
12.47	8.98	none	incipient	0.08
13.53	9.10	sustained	incipient	0.03
14.34	8.53	sustained	incipient	0.18
14.99	7.79	incipient	sustained	0.02

documented g-break followed by pitch bucking, a slow oscillation as the tail regains power at each nose drop, with no wing drop; its g-break of 1.1–8.0° at the aft loading is about three times the reported 2–3°. FG shows no break at its two aft loadings. In the dynamic entry both models reach the reported ≈ 20 KIAS (20–27 and 21–28 kt), but v9 drops the nose by 65–78° against the reported "up to 40"; height loss (166–242 ft for v9, 149–167 ft for FG, against ≈ 100 ft) depends on the scripted recovery. The inverted stall separates the models clearly: v9 gives 40.6 KIAS (AFFTC 38–40), FG 54.4 KIAS, because its trimmed negative lift peaks at $C_L = -0.53$.

7.5 Spin susceptibility

This is the central result (Fig. 9). FG does not spin at any of the 24 loadings; it agrees with the AFFTC only at the 2 loadings forward of the incipient boundary, where no spin is expected. With full pro-spin controls it settles in a spiral dive at α of 11–20°, bank of 50–75°, increasing speed and up to 7 g when aileron is held into the turn (Fig. 10). The cause is structural: its C_L stays at 0.90 beyond 25° and its roll damping $C_{l_p} = -0.4$ is constant at every α, so a stalled wing never produces the autorotative rolling moment that drives a spin, and its inertias are too small for the inertial pitching moment

Table 8: Developed spins: v9 statistics over all sustained spins (from 12 s after the stall, all entries) against the AFFTC description.

Developed spin	AFFTC-TR-89-27	FlightGear	v9 (all entries)
sustained spins in the programme	–	0 of 72	27 of 72
rotation rate (deg/s)	80 (flat)–140 (steep)	–	104 (79–128)
time per turn (s)	2.5–4.5	–	3.5 (2.8–4.5)
height loss per turn (ft)	200 (150–250)	–	291 (240–365)
mean pitch, low inertia (deg)	−50 ± 15	–	−37, σ = 19
mean pitch, high inertia (deg)	−40 ± 30	–	−39, σ = 20
mean rate low / high inertia (deg/s)	higher at high I	–	100 / 109
angle of attack (deg)	stalled, "fast, steep"	–	50 (44–55)
airspeed (KIAS / model CAS)	30–40 (indicated)	–	48 (41–55)
normal load factor	≈ 1.0	–	1.39 (1.16–1.87)

that the AFFTC identified as important. v9 classifies 20 of 24 loadings as the AFFTC did. Of the 4 differences (Table 7), 2 are one class too spin-resistant and 2 one class too spin-prone; 4 lie within 0.2 in of a boundary. Of the 16 loadings where the AFFTC obtained sustained spins, v9 sustains a spin with the plain wings-level entry at 10, with the thermal entry at 10 and with aileron held into the spin at 6 (Appendix A). The AFFTC found the thermal entry the more successful technique and aileron into the spin to make spins more sustainable.

7.6 Developed spin

v9's developed spin is erect, fast and oscillatory, the AFFTC's modifiers (Fig. 10, Table 8). Over its 27 sustained spins the mean rate of 104 deg/s and 3.5 s per turn lie inside the reported ranges, and the mean pitch attitude of −38° and mean α of 50° match "steep". The oscillation is larger than reported: the pitch attitude ranges from −88° to 3° against ±15° about −50° (low inertia) or ±30° about −40° (high inertia). The mean rate is 100 deg/s at low and 109 deg/s at high inertia; the AFFTC found the highest average rates at forward CG and high inertia. Height loss per turn (291 ft) exceeds the

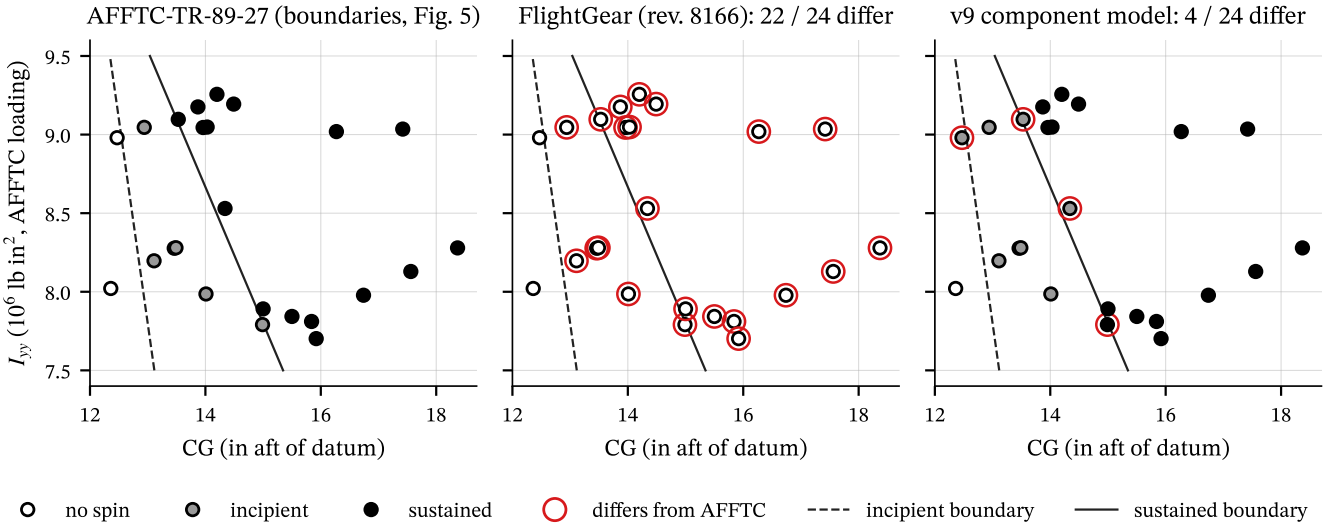


Figure 9: Spin susceptibility over the 24 AFFTC loadings, plotted at the AFFTC's CG and I_{yy}; boundaries from Eqs. (1)–(2). Each model loading is classified by its most spin-prone entry (WL, WL+A, TH).

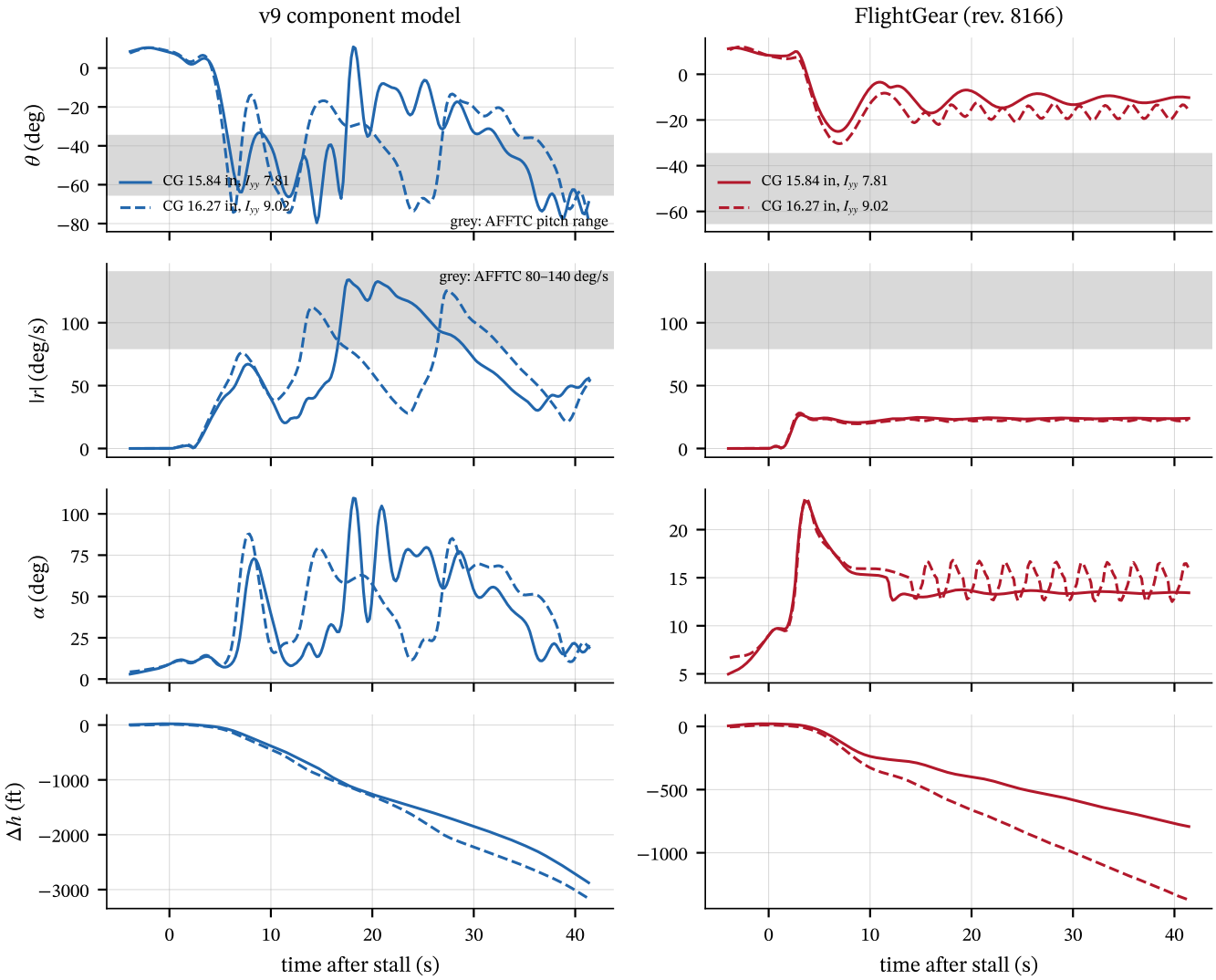


Figure 10: Thermal-entry spins at a low-inertia (15.84 in) and a high-inertia (16.27 in) loading. Grey bands: AFFTC pitch attitude (low inertia) and rotation-rate ranges. FG (right) spirals at 20–25 deg/s with α below 25°.

AFFTC's 200 ft, and the model's mean airspeed (48 KCAS) exceeds the indicated 30–40 KIAS.

How consistent is "200 ft per turn"? With a near-vertical flight path the height loss per turn is $\Delta h = 2\pi V_z/\Omega$. Equilibrium requires $W = \frac{1}{2}\rho V_z^2 S\, C_v(1+\kappa)$, where $\kappa = \Omega^2 \int c\, y^2\, \mathrm{d}y/(SV_z^2)$ is the extra dynamic pressure that rotation brings to the outer wing ($\int c\, y^2\, \mathrm{d}y = 327\,\mathrm{m}^4$). For the AFFTC's nominal values (200 ft at 105 deg/s, 8500 ft altitude,

1150 lb) these give V_z = 17.8 m/s TAS (30 kt EAS, consistent with the indicated 30–40 KIAS), $\kappa = 0.19$, and a required vertical-force coefficient $C_v = 1.59$. That is at or above the two-dimensional flat-plate normal force at the spin incidence ($1.98 \sin\alpha$ = 1.40–1.52 at 45–50°), and above any three-dimensional wing value. v9 develops $C_v = 0.78$; its static resultant coefficient at 45–50° is 1.04–1.13. Either the real spin was flatter than its reported attitude, or the altimeter-based height loss is biased low; a pressure-altimeter lag of a few seconds, for instance, would under-read the height lost over a three-turn spin by about 150 ft at a 17 m/s descent, i.e. 50 ft per turn. The physically plausible value probably lies between 200 and 300 ft per turn. Within the present model structure the identification did not find a parameter set that matches the height loss together with the spin map: it settled on a post-stall normal force of 1.15 and a fuselage cross-flow drag coefficient of 1.19.

7.7 Recovery

The flight-manual procedure (opposite rudder, a short pause, stick forward) recovers v9 in 0.53–0.76 turns, losing 323–340 ft until the rotation stops (AFFTC: 200–300 ft). Opposite rudder alone, with the stick held aft, recovers in 0.63–1.05 turns, against the AFFTC's $\frac{1}{4}$–$\frac{3}{4}$ typical and $1\frac{1}{2}$ maximum (Fig. 11, Table 9). Holding the aileron against the spin fails to recover at 3 of the 3 loadings, which agrees with "not a reliable contributor". Forward stick with pro-spin rudder recovers v9 after 1.11–1.21 turns; the AFFTC saw up to three additional turns, so v9 is somewhat too benign in this respect. At the high-inertia 14.20 in loading the spin develops more slowly than the 14 s the test allows before the input (it is sustained in the spin tests, which hold the controls for 40 s), so its recoveries are not tested there. FG has no spin from which to recover.

7.8 Lateral–directional handling

The 45–45° reversal takes 4.5–5.7 s in v9 and 4.5–5.2 s in FG with and without rudder (Fig. 12); no measured value is available. A full-aileron roll at the manual's slow-roll entry speeds (165 km/h dual, 150 km/h solo) reaches 180° after 4.5–4.8 s in v9 and 6.2–6.6 s in FG, against ≈ 5 s in the flight video. In sideslip v9 reaches 46° with full rudder (AFFTC: up to 50°) and sinks at 9.1 m/s, 1.6 times the 5.6 m/s of a full-airbrake glide at the same airspeed. FG's linear side-force and yaw derivatives cap the slip at 16°, where it sinks at 3.6 m/s, less than with airbrakes (4.8 m/s). The rudder lock-out reported in large slips needs a hinge-moment model and is outside both models.

7.9 Ground attitude and take-off

On its wheels v9 rests at 0.5–0.6° pitch (nose- and main-wheel contacts corrected to the manufacturer's drawing, Section 3), FG at −0.9 to −0.8°. With the stick centred, a glider lifts off at roughly its stick-free trim speed plus a few km/h (Fig. 14, Table 11). v9 lifts off at 66–80 km/h for normal loadings, before the tug and close to the manual's "about 75 km/h"; two heavy pilots at the forward limit need a slight pull (91 km/h hands-free). FG lifts off at 91–108 km/h, after the tug, and with the heavy forward loading not at all. The ground-attitude correction alone advances the hands-free lift-off by only 1–5 km/h: the nose wheel comes up when the pitching moments balance, which depends on the stick-centred elevator angle rather than on the wing's angle on the ground. With the mid-travel elevator angle (−3.2°) two 80 kg pilots would lift off at 88 km/h, after the tug.

7.10 Small-perturbation modes

The modes (Fig. 15, Table 12) show the consequences of FG's mass and damping data. Its roll subsidence time constant is 0.09 s against 0.35 s for v9 with the measured I_{xx}, and its Dutch roll is a stiff 2.9 s oscillation (v9: 7.6 s). v9's damping derivatives are close to the classical estimates for this tail volume, $C_{m_q} \approx -2a_t\eta V_H l_t/\bar{c} \approx -16$ for the tail alone (v9 −18.5 in total) and $C_{m\dot{\alpha}} \approx C^t_{m_q}\,\mathrm{d}\varepsilon/\mathrm{d}\alpha \approx -4.5$ (v9 −3.6); FG's −10.6 is more than twice that estimate. The phugoid is the one mode where v9 is worse (Table 13). Its damping ratio is negative at low speed, −0.083 at 80 km/h (the amplitude doubles within about two cycles) and −0.019 at 90 km/h, and turns positive between 110 and 120 km/h (−0.001 at 110, +0.033 at 140 km/h); FG's is +0.076 to +0.114. Both periods exceed Lanchester's $\pi\sqrt{2}\,V/g$ (11.3 s at 90 km/h) because of the coupling with the pitch damping: without M_q and $M_{\dot{\alpha}}$ both linear models return to about 11.5 s. For a glider of this glide ratio the classical damping is $\zeta \approx 1/(\sqrt{2}\,L/D) \approx 0.02$, a small margin that the derivative sensitivity analysis shows to depend mainly on M_w, M_q and $M_{\dot{\alpha}}$; it gives no single unphysical cause for the negative value. Measured phugoid damping for the ASK 21 is not available, but a glider that diverges in pitch within two oscillations at 80 km/h would be noticed; this is the most consequential open defect of v9 for normal flying.

7.11 Robustness

Both models were parked for 30 s in 15, 25 and 35 kt wind from five directions between head- and tailwind, and flown through 60 random upsets ($|\omega| \le 3$ rad/s, α and attitude anywhere, random controls and wind). All parked runs stayed finite, with pitch excursions of up to 5.5° (FG) and 0.8° (v9) while the glider settled onto a wingtip; 0 (FG) and 0 (v9) upset runs produced a non-finite state.

8 Summary scorecard

Table 14 condenses the comparison. Of 24 criteria, FG meets 2, is qualitatively right in 4 and fails 18; v9 meets 17, is qualitatively right but quantitatively off in 6 and fails 1.

9 Changes since the first issue

Third issue. The model changed in three ways, and its post-stall factors were re-identified (Section 4):

- **Wing bending.** Each panel's first bending mode, with the stiffness of the KTH ground vibration test (Section 4).
- **Rate.** The model runs at 240 Hz instead of 120 Hz: at 120 Hz the integration error decided some borderline spins.
- **Property reads.** The function graph was laid out for FlightGear's property locks (Section 4). The forces and moments stay within 3×10^{-7} of the earlier layout's (the constants are written to six digits, which alone moves them by 3×10^{-6}),

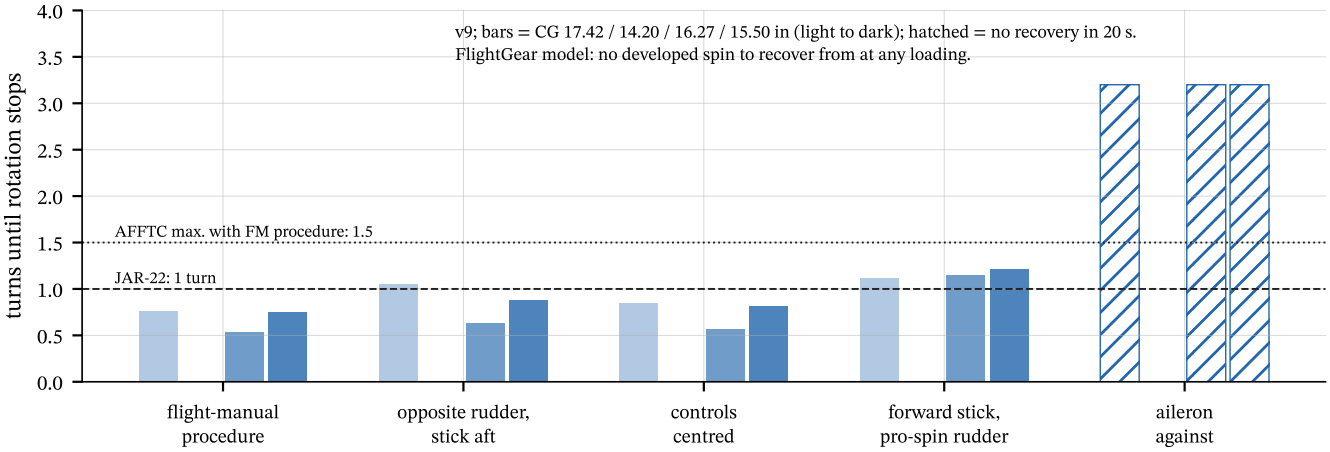


Figure 11: v9: turns from the recovery input until the rotation stops.

Table 9: v9 recoveries: turns (height loss) until the rotation stops, at four AFFTC loadings (CG in / I_{yy}). n.s. = not spinning at the time of the input; none = no recovery within 20 s. [a] The AFFTC released the controls ("hands off"); a free-floating control model would be needed for that test.

v9 input \ loading	17.42/9.04	14.20/9.26	16.27/9.02	15.50/7.84	AFFTC
flight-manual procedure	0.76 (340 ft)	n.s.	0.53 (339 ft)	0.75 (323 ft)	< 1, max. $1\frac{1}{2}$
opposite rudder, stick aft	1.05 (503 ft)	n.s.	0.63 (365 ft)	0.88 (371 ft)	$\frac{1}{4}$–$\frac{3}{4}$, max. $1\frac{1}{2}$
all controls centred	0.85 (373 ft)	n.s.	0.56 (313 ft)	0.82 (325 ft)	not tested[a]
full forward stick, pro-spin rudder	1.11 (339 ft)	n.s.	1.15 (313 ft)	1.21 (305 ft)	up to 3 more turns
aileron against, stick aft	none	n.s.	none	none	unreliable
pro-spin controls held	none	n.s.	none	none	spin continues

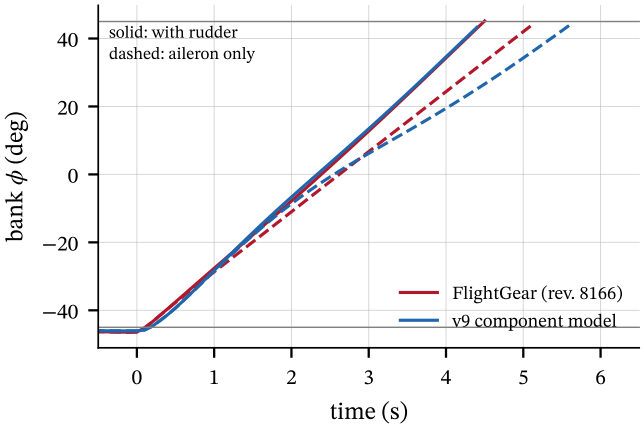


Figure 12: 45°–45° roll reversal at 90 km/h, 2 × 80 kg, full aileron.

Table 10: Roll, sideslip and airbrakes. The airbrake sink is taken from a trimmed glide at the airspeed of the slip. [b] Manual: "sideslipping is also suitable as an approach control"; the commissioning pilot's experience is that a full slip is at least as effective as the airbrakes.

	Reference	FlightGear	v9
45°–45° reversal, 90 km/h, with / without rudder (s)	–	4.54 / 5.22	4.52 / 5.71
full-aileron roll, dual, 165 km/h: 180° / 360° (s)	≈ 5 / –	6.2 / 10.1	4.5 / 7.5
full-aileron roll, solo, 150 km/h: 180° / 360° (s)	≈ 5 / –	6.6 / 11.7	4.8 / 8.0
full-rudder sideslip, 2 × 80 kg: β (deg)	up to 50	16	46
slip flown at (km/h CAS)	–	88	99
sink in the slip / full airbrakes at the same speed (m/s)	slip ≳ brakes[b]	3.6 / 4.8	9.1 / 5.6

and the whole test programme at 120 Hz gave the same results to the printed precision.

- **Test programme.** The polar sweep now runs down from high speed: with the bending the 470 kg sweep, which started below the stall, settled on the stalled branch. The identification's stall-speed check had used an outdated copy of the aircraft file.

Table 11: Stick-free trim speed (trim neutral) and hands-free aerotow lift-off. [c] FG has no trim point below the stall with the stick centred at these CGs.

Loading (front / rear, lb)	stick-free trim (km/h)	hands-free aerotow lift-off (km/h)		
	FG / v9	FlightGear	v9	tug (85 km/h)
176 / 176 (CG 12.3 in)	none[c] / 73	108	80	v9 first
176 / 0 (CG 15.1 in)	none[c] / 63	91	66	v9 first
230 / 0 (CG 11.7 in)	none[c] / 70	105	77	v9 first
230 / 230 (CG 9.0 in)	none[c] / 89	–	91	tug first

Second issue. The first issue contained errors that a line-by-line check against the code and the results files revealed. They are listed here because three of them changed the model.

- **Fuselage potential load.** The slender-body (Munk) term multiplied the *total* dynamic pressure, including the sideslip component, by $\sin 2\alpha$ of the flow in the vertical plane (and vice versa laterally). This inflates the load by $V^2/(u^2+w^2)$: by up to ≈ 20 % in spins, and without bound in pure cross-flow, where it tipped the parked glider onto its nose in a 15 kt crosswind gust. The corrected term is $\rho\, u\, w\, \Delta S$ (Section 4).
- **Force direction.** Section forces were directed normal to the 3/4-chord flow instead of the 1/4-chord flow (bound vortex), which produced a spurious $X_q \approx +18$ lbf s/rad.
- **Nose-gear contact.** See Section 3; the gear inherited from the FlightGear model rested the glider ≈ 1.4° too far nose-down.
- The 18 parameters were re-identified after these corrections (Section 4).

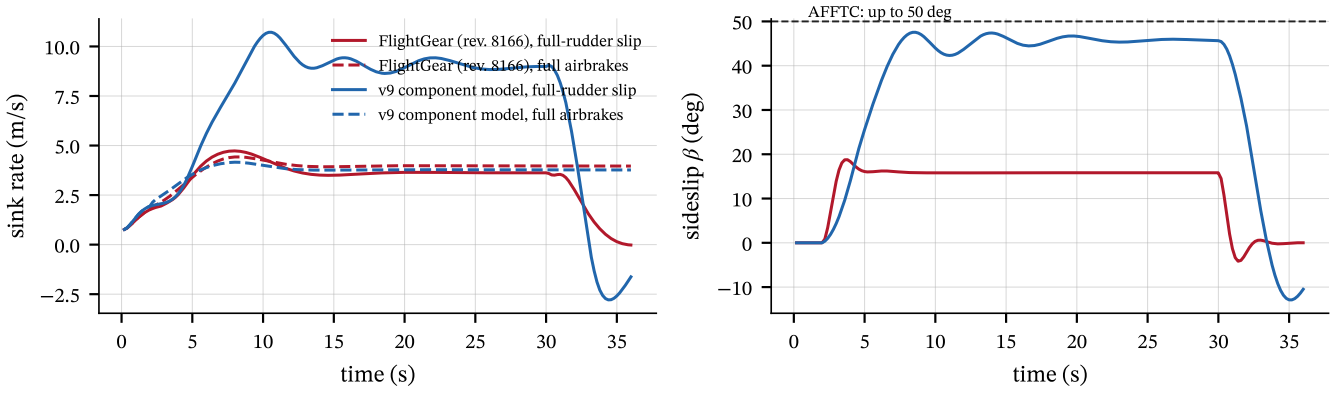


Figure 13: Full-rudder sideslip with the heading held by bank (target 90 km/h; the slips settle at 88 and 99 km/h CAS), and a straight glide with full airbrakes.

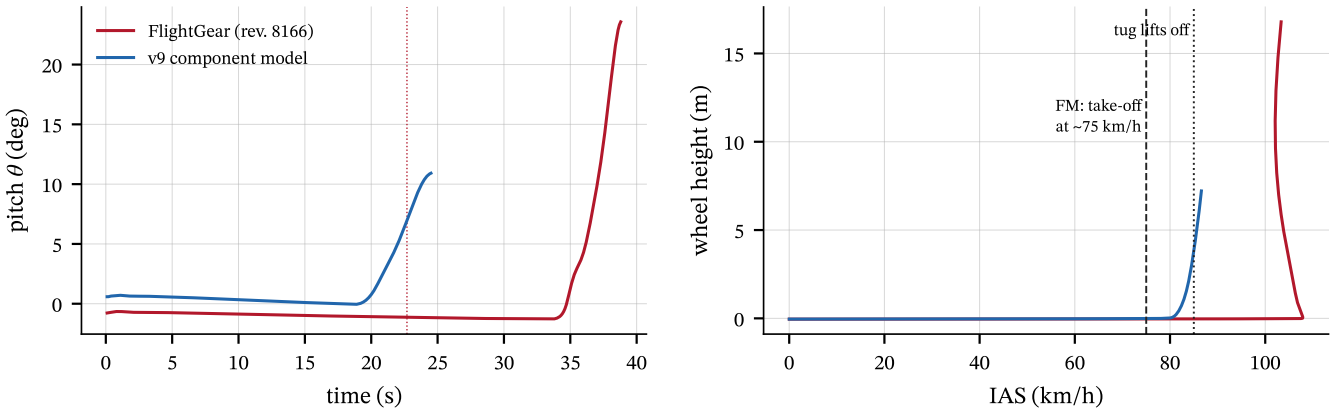


Figure 14: Aerotow, 2 × 80 kg, stick centred and trim neutral: 40 m rope to a tug that accelerates at full power and lifts off at 85 km/h.

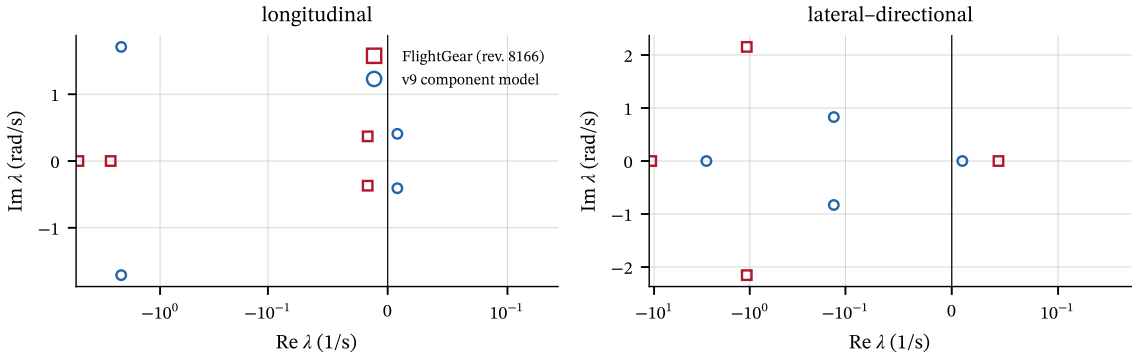


Figure 15: Eigenvalues about a trimmed 90 km/h glide (2 × 80 kg); symmetric-log real axis.

- **Test definitions.** The 45° airbrake dive had been flown at a 45° pitch attitude (a 40° flight path); the stall speed had been taken at the α peak, which for FG at aft CG lies after its lift has collapsed and had suggested a stall speed rising toward aft CG; the slip had been compared with an airbrake glide at a different airspeed; the parked-in-wind test had not been run for FG; and the phugoid had been characterised at a single speed.
- **Text.** Several statements had not been checked against the final results: the direction of the spin-map misclassifications, the effect of the entry technique, the ground attitude (stated as 1.8° nose-down), the attribution of informal reports, and the document titles of two sources.

10 Discussion and limitations

- **Why FG cannot spin.** Its C_L table is clamped at 0.90 beyond 25° and its roll and yaw damping are constants, so no combination of α and rate produces a pro-spin rolling moment. Its inertias are 68–85 % too small, so the inertial pitching moment that holds a real spin at a stalled α is missing. These are structural defects, not tuning errors.
- **What is fitted and what is predicted.** The 18 identified factors were fitted to the spin map, the spin characteristics, the recoveries, four 1-g stalls and the manual's stall speeds. Earlier, two drag factors (profile-drag factor 1.00, miscellaneous C_D 0.0017) were set from the manual polar and the airbrake drag from the manual's dive speed. The stick-centred elevator angle was chosen for the hands-free take-off, and the wing's stiffness taken from the ground vibration test. Not fitted to anything, and therefore predictions, are the roll response, the sideslip capability and the slip sink

Table 12: Non-dimensional derivatives (per rad; $\bar{c} = 1.121$ m, $b = 17$ m, rates normalised by $\bar{c}/2V$ and $b/2V$) and modes from the linearised models. τ = time constant, t_2 = time to double amplitude, $t_{1/2}$ = time to halve it.

Derivative	FlightGear	v9	Mode	FlightGear	v9
C_{L_α}	+6.863	+6.303	phugoid	17.0 s, ζ 0.045	15.4 s, ζ −0.020
C_{m_α}	−0.680	−1.412	short period	overdamped: τ 0.17, 0.35 s	3.7 s, ζ 0.80
C_{m_q}	−11.949	−18.505	roll subsidence τ	0.09 s	0.35 s
$C_{m_{\dot\alpha}}$	−10.622	−3.613	Dutch roll	2.9 s, ζ 0.447	7.6 s, ζ 0.157
C_{L_q}	−0.000	+8.983	spiral	t_2 16 s	t_2 70 s
C_{Y_β}	−1.028	−0.330			
C_{l_β}	−0.106	−0.115			
C_{n_β}	+0.149	+0.034			
C_{l_p}	−0.400	−0.676			
C_{n_p}	−0.000	−0.080			
C_{l_r}	+0.150	+0.194			
C_{n_r}	−0.150	−0.044			
trim: 90 km/h, 2 × 80 kg, CG 12.1 / 12.3 in; static margin $-C_{m_\alpha}/C_{L_\alpha}$ = 9.9 / 22.4 % MAC ⇒ x_{NP} = 16.5 / 22.1 in					

Table 13: Phugoid from nonlinear simulation (1.5 m/s speed disturbance, stick fixed, 2 × 80 kg, CG 12.1–12.3 in).

V (km/h)	80	90	100	110	120	140
FlightGear: period (s)	–	16.9	19.0	21.1	23.3	29.8
damping ratio ζ	–	+0.076	+0.064	+0.056	+0.056	+0.114
v9: period (s)	13.3	15.4	17.1	18.9	20.6	24.1
damping ratio ζ	−0.083	−0.019	−0.010	−0.001	+0.009	+0.033

rate, the airbrake stall-speed increment, the inverted stall, the static and dynamic stability derivatives and the phugoid.

- **Sensitivity of the boundary.** Near the boundary, outcomes change with the entry details: whether the stall coincides with peak yaw rate, the elevator gearing, the pause length. The AFFTC observed the same ("spin entry success was sensitive to entry pitch attitude conditions"). Classifications within ≈ 0.2 in of a boundary should be read as ties.
- **Not modelled.** Free-floating controls and hinge moments (hands-off spin behaviour, rudder lock-out in sideslip, stick-free stability); pitot–static errors (all model speeds are CAS); unsteady stall hysteresis beyond the downwash lag; the wing's torsion and its higher bending modes (the first bending mode is modelled, and FlightGear's 3-D wings bend with it).
- **Open quantitative gaps in v9.** The divergent phugoid below about 115 km/h; spin height loss and oscillation amplitude; an aft-CG g-break about three times the reported value; too large a nose drop in dynamic stalls; and a spin that develops too slowly at the high-inertia 14.20 in loading to be recovered from in the recovery test.
- **Assumed values.** The stick-centred elevator angle (−6°), the 85 km/h tug lift-off speed and the 1 lb tail-weight bracket are assumptions; the first is justified only by the manual's take-off statement and the commissioning pilot's experience.

11 Conclusions

A lumped-coefficient model of Aeromatic origin, even one tuned to look right in cruise, cannot reproduce the stall and spin behaviour of a training sailplane. In the case studied, the FlightGear reference model glides 25 % too well, is statically unstable at aft CG, and turns every spin attempt into a spiral dive. A component build-up from public geometry, airfoil and mass data, closed by a small number of physically bounded post-stall factors, does markedly better. It matches the manual's polar within the reading accuracy of the chart and the manual's stall speeds within a few km/h, reproduces 20 of the 24 AFFTC spin classifications including their dependence on pitch inertia, and reproduces the documented recovery characteristics, the large sideslip capability and the hands-free take-off. Its remaining errors are quantified: the most consequential for normal flying is a phugoid that diverges at low speed; the largest against the spin data is the height loss per turn, which sits next to an apparent inconsistency in the reference data itself.

A Per-loading spin outcomes

Table 15 lists every spin test point.

B Reproducibility

The model generator (`genxml.py`, `genjsb.py`, `airfoils.py`, `liftingline.py`, `geometry.py`, `flexwing.py` for the wing bending), the identification (`objective.py`, `optimize.py`, `optimize2.py`; the environment variable `FDM_HZ` sets the rate, 240 for v9) and the test campaign with its evaluation (`campaign.py`, `summarize.py`, `figures.py`, `mktables.py`, `mknumbers.py`, `paradox.py`) are shipped with the model. `campaign.py <model>` flies the 186 test points of this report; every number in the text and tables is taken from its output by `mktables.py` and `mknumbers.py`, except the reference values and the verdicts of the scorecard. The identified parameter set is `v9_params.json`.

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Table 14: Behaviour against the sources. ✓ within the reported range or tolerance; ~ qualitatively right, quantitatively off; × wrong or absent.

Criterion	Source	Reference	FlightGear rev. 8166		v9	
Glide polar, rms sink error	FM p. 55	–	0.09–0.17 m/s	×	0.025–0.030 m/s	✓
Best glide ratio	FM p. 55	33.5	42.3–42.5	×	32.6–32.7	✓
Stall speed single / dual	FM	65 / 74 km/h	60.3 / 69.3	×	63.9 / 72.8	✓
Stall-speed increase, airbrakes	FM, AFFTC	3 km/h, 2 KIAS	1.0–1.1	×	2.4–2.7	✓
Airbrake 45° dive, 600 kg	FM	232 km/h	194	~	232	✓
Neutral point aft of the aft CG limit	–	> 18.46 in	16.5 in	×	22.1 in	✓
1-g stall: forward CG elevator-limited	AFFTC	yes	yes	✓	yes	✓
1-g stall: aft CG g-break + bucking	AFFTC	2–3°, bucking	0.0–4.0°	~	1.1–8.0°	~
Dynamic stall: min. speed / nose drop	AFFTC	20 KIAS / ≤40°	21–28 / 0–55°	~	20–27 / 65–78°	~
Inverted −1g stall	AFFTC	38–40 KIAS	54.4	×	40.6	✓
Spin map, 24 loadings	AFFTC Fig. 5	–	2 of 24	×	20 of 24	✓
Thermal entry more effective than wings-level	AFFTC	yes	0 vs 0	×	10 vs 10	~
Spin rate (mean)	AFFTC	80–140 deg/s	no spin	×	104	✓
Spin pitch attitude / oscillation	AFFTC	−50 ± 15° (low I)	–	×	−37, σ = 19°	~
Height loss per turn	AFFTC	200 (150–250) ft	–	×	291	~
Recovery, FM procedure (turns)	FM, AFFTC	< 1 (max 1$\frac{1}{2}$)	–	×	0.53–0.76	✓
Recovery, opposite rudder only (turns)	AFFTC	$\frac{1}{4}$–$\frac{3}{4}$ (max 1$\frac{1}{2}$)	–	×	0.63–1.05	✓
Forward stick, pro-spin rudder (turns)	AFFTC	up to 3 more turns	–	×	1.11–1.21	~
Full-rudder sideslip	AFFTC	up to 50°	16°	×	46°	✓
Slip sink vs. airbrakes, same speed	FM, pilot	slip ≳ brakes	3.6 vs 4.8 m/s	×	9.1 vs 5.6 m/s	✓
Full-aileron roll, first 180°	video	≈ 5 s	6.2–6.6 s	~	4.5–4.8 s	✓
Hands-free aerotow lift-off before the tug	FM, pilot	≈ 75 km/h	91–108 km/h	×	66–80 km/h	✓
Empty inertias	AFFTC App. C	30.2 / 6.7 / 34.3	4.6 / 1.4 / 10.9	×	30.2 / 6.7 / 34.3	✓
Phugoid damped at 80–140 km/h	theory	$\zeta \approx 0.02$	min. ζ = 0.06	✓	min. ζ = −0.08	×

Rules: polar rms < 0.05 m/s (~ < 0.10); best glide 31.5–35 (~ 30–37); stall speeds within 2 km/h (~ 4); airbrake increment 2–4.5 km/h; dive within 5 % (~ 20 %); inverted stall 36–42 KIAS (~ 32–46); spin map ≥20 of 24 (~ ≥16); spin rate 80–140; pitch mean −35 to −60° with $\sigma \le 12°$; height loss 150–250 ft (~ ≤350); recoveries ≤ 1$\frac{1}{2}$ turns; forward stick recovering at all is too benign (~); sideslip ≥ 35° (~ ≥ 25°); roll 4–6 s (~ 3–7); lift-off below the tug's 85 km/h with 2 × 80 kg; inertias within 5 %; phugoid $\zeta > 0$ at all tested speeds. Stall-break rows: ✓ as described, ~ bucking present / break < 3.5°.

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Table 15: All 24 AFFTC loadings: pilot masses (lb), number of tail / cockpit ballast weights, I_{yy} (10^6 lb in^2), the AFFTC expectation (Eqs. 1–2) and each model's outcome per entry technique (I = incipient, S = sustained, small numbers = spinning turns within 40 s; – = no spin).

AFFTC Table A1 loading						FlightGear			v9		
CG	front	rear	tail/ckpt	I_{yy}	expected	WL	WL+A	TH	WL	WL+A	TH
12.36	176	230	0/0	8.02	–	–	–	–	–	–	–
12.47	230	176	8/6	8.98	–	–	–	–	–	I 4.6	I 3.9
12.94	230	176	9/5	9.05	I	–	–	–	–	I 4.1	I 4.7
13.11	176	230	2/0	8.20	I	–	–	–	–	I 4.9	–
13.46	176	230	3/0	8.28	I	–	–	–	–	I 4.4	I 1.0
13.49	176	217	3/1	8.28	I	–	–	–	–	I 4.4	–
13.53	230	176	10/3	9.10	S	–	–	–	–	I 4.2	I 6.1
13.87	230	176	11/3	9.18	S	–	–	–	–	I 4.1	S 7.7
13.96	176	230	10/14	9.04	S	–	–	–	–	I 3.9	S 7.4
14.01	176	0	0/11	7.99	I	–	–	–	–	I 4.0	–
14.03	217	176	10/4	9.05	S	–	–	–	–	I 4.0	S 7.7
14.20	230	176	12/3	9.26	S	–	–	–	S 6.1	I 4.2	S 7.2
14.34	176	230	6/1	8.53	S	–	–	–	–	I 3.9	I 6.1
14.49	217	230	12/0	9.19	S	–	–	–	S 6.7	S 8.3	S 7.7
14.99	139	176	0/0	7.79	I	–	–	–	I 2.2	S 7.4	–
15.00	176	0	0/3	7.89	S	–	–	–	S 5.9	S 7.3	I 4.0
15.50	176	0	0/2	7.84	S	–	–	–	S 5.5	S 7.3	I 3.7
15.84	176	0	0/0	7.81	S	–	–	–	I 1.6	S 7.3	I 6.8
15.92	122	176	0/0	7.70	S	–	–	–	S 7.1	S 7.3	I 6.1
16.27	176	217	12/3	9.02	S	–	–	–	S 7.3	S 5.5	S 6.8
16.74	176	0	2/0	7.98	S	–	–	–	S 5.9	–	S 7.6
17.42	230	0	12/0	9.04	S	–	–	–	S 6.4	–	S 6.9
17.56	176	0	4/0	8.13	S	–	–	–	S 6.8	–	S 6.5
18.37	176	0	6/0	8.28	S	–	–	–	S 5.4	–	S 7.0